06

Patterns, expressions and equations

Part III · Patterns and symbols

CHAPTER 6 · OPENER

Patterns, expressions and equations

A pattern can be described by a rule; expressions, equations and formulas make the relationship precise and allow prediction and design.

ENGAGEMENT

How can a gardener order exactly the right number of border slabs for a square pond of any whole-number side length?

Learning goals

- Describe and generalise a growing pattern.
- Substitute into expressions and simplify or expand.
- Solve linear equations and check solutions.
- Use formulas and budget constraints in a design.

MYP lens

Key concept: Relationships

Related concepts: Pattern, Equivalence

Global context: Scientific and technical innovation

ATL focus. In order for a student to transfer mathematical knowledge, students must move between a diagram, a table, a rule and an equation, then check the result in context.

Language bank: sequence · term · position · variable · expression · coefficient · like terms · expand · factorise · equation · formula · solution

CHAPTER 6 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the chapter map from concrete examples to explanations and decisions.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

R1	Calculate 3×4+5.	READY
R2	Continue 2, 5, 8, 11 by one term.	READY
R3	Find 4×7 and 28÷4.	READY
R4	Simplify 2x+3x.	READY
R5	What number added to 7 gives 15?	READY

Chapter map

1. **6.1 Patterns, rules and sequences** — Describe how a pattern grows, then connect the picture to a rule for any position.
2. **6.2 Expressions and substitution** — A letter stands for a number. Replace it carefully and follow the order of operations.
3. **6.3 Simplifying and expanding expressions** — Combine only like terms; use the distributive law to remove brackets.
4. **6.4 Equations, formulas and checking solutions** — An equation states equality. Keep both sides balanced, then substitute the solution to check.

> **EXPLAIN HOW YOU KNOW**
>
> For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 6 · MATHEMATICAL DEVELOPMENT

6.1 Patterns, rules and sequences

BIG IDEA

Describe how a pattern grows, then connect the picture to a rule for any position.

A sequence has terms in positions 1, 2, 3, ... A term-to-term rule says how to get the next term. A position-to-term rule uses the position n to find a term directly.

For 4, 7, 10, 13, ..., add 3 each time. The nth term is 3n + 1 because position 1 gives 4. A rule must match every given term, not only the first.



1 square: 4 matches **2 squares: 7 matches** **3 squares: 10 matches**

Each extra square shares one side and adds 3 matches.

Each new square shares a side: 4, 7, 10 matches. The rule is 3n + 1.

WORKED EXAMPLE · A GROWING MATCHSTICK CHAIN

One square uses 4 matches; each extra square shares one side and adds 3.

1. n squares use 4 + 3(n−1) matches.
2. Simplify to 3n + 1.
3. For 10 squares: 31 matches.

3n + 1; 31 matches for 10 squares.

CHAPTER 6 · PRACTICE

6.1 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

A pattern in the first few terms does not prove a unique rule. Explain the structure that makes the rule continue.

GUIDED PRACTICE · FIRST STEPS

6.1.1 Continue 4, 7, 10, 13 by two terms and state the term-to-term rule. FLUENCY

6.1.2 Find the nth term of 4, 7, 10, 13, ... and the 20th term. FLUENCY

6.1.3 Explain why n squares in a shared-side chain need 3n + 1 matches. REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

6.1.4 A square pond of side n metres has a one-slab border. Find the border rule and the count for n=7. APPLICATION

6.1.5 Does 50 matches make a whole number of squares in the 3n+1 chain? REASONING

6.1.6 What is the largest whole number of squares that can be made with 100 matches in this chain? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 6 · MATHEMATICAL DEVELOPMENT

6.2 Expressions and substitution

BIG IDEA

A letter stands for a number. Replace it carefully and follow the order of operations.

An expression such as 3n + 5 combines numbers, letters and operations. It has no equals sign. A term is a part added or subtracted; a coefficient multiplies a letter.

To evaluate, substitute the value and use brackets where needed. If n=4, 3n+5 = 3×4+5 = 17. If x=−2, $x^2 = 4$ but $-x^2 = -4$.

WORKED EXAMPLE · EVALUATE 2A + 3B

Let a=5 and b=2.

1. 2×5 + 3×2.
2. 10 + 6.

16.

CHAPTER 6 · PRACTICE

6.2 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

3n means 3 × n, not 3 + n. Write the multiplication sign when substituting to avoid losing it.

GUIDED PRACTICE · FIRST STEPS

6.2.1 Evaluate $3n+5$ for $n=4$ and $n=10$. FLUENCY

6.2.2 Evaluate $2a+3b$ for $a=5$, $b=2$. FLUENCY

6.2.3 Evaluate x^2 and $-x^2$ for $x=-2$. Explain the difference. REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

6.2.4 A taxi charges £3 plus £2 per km. Write the cost for d km and evaluate for 8 km. APPLICATION

6.2.5 A student says $4n=4+n$ when $n=3$. Correct the statement. REASONING

6.2.6 Find n if $2n+7=25$, then check by substitution. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 6 · MATHEMATICAL DEVELOPMENT

6.3 Simplifying and expanding expressions

BIG IDEA

Combine only like terms; use the distributive law to remove brackets.

Like terms have the same letter part. 3x + 5x = 8x, but 3x + 5 cannot be combined into one term. Rearranging terms does not change their value.

Expanding uses a(b+c)=ab+ac. For 3(2x+4), multiply every term inside: 6x+12. Factorising reverses this: 6x+12 = 3(2x+4).

WORKED EXAMPLE · SIMPLIFY 4X + 3 + 2X − 5

Group like terms.

1. 4x+2x = 6x.
2. 3−5 = −2.

6x − 2.

CHAPTER 6 · PRACTICE

6.3 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

$x + x = 2x$, not x^2. $x \times x = x^2$. Addition and multiplication are different operations.

GUIDED PRACTICE · FIRST STEPS

6.3.1 Simplify $4x+3+2x-5$ and $7a-2a+4$. FLUENCY

6.3.2 Expand $3(2x+4)$ and $5(y-2)$. FLUENCY

6.3.3 Explain why $x+x$ is not x^2. REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

6.3.4 A rectangle has length $2x+3$ and width 4. Write and expand its perimeter. APPLICATION

6.3.5 Is $3(x+2)=3x+2$ true for every x? Test and correct it. REASONING

6.3.6 Factorise $12x+18$ completely and check by expanding. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 6 · MATHEMATICAL DEVELOPMENT

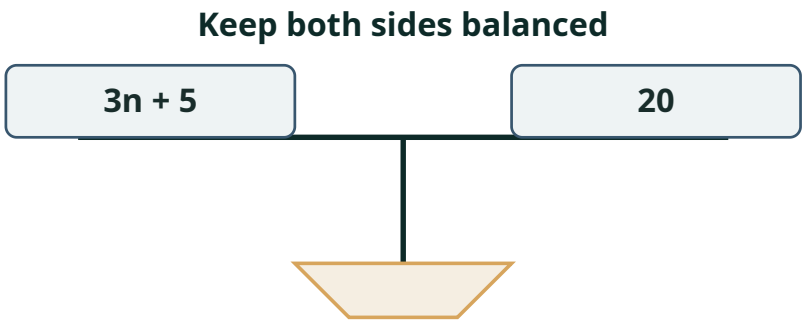
6.4 Equations, formulas and checking solutions

BIG IDEA

An equation states equality. Keep both sides balanced, then substitute the solution to check.

To solve, undo operations in reverse order. For 3n+5=20, subtract 5 from both sides, then divide by 3: n=5. Whatever you do to one side, do to the other.

A formula connects quantities, such as C=3+2d. To find d when C=19, solve 19=3+2d. A solution should satisfy the original equation and make sense in context.



Subtract 5 from both sides, then divide both sides by 3: n = 5.

WORKED EXAMPLE · SOLVE 4N + 4 = 44

Undo addition, then multiplication.

1. 4n = 40.
2. n = 10.
3. Check: 4×10+4 = 44.

n=10.

CHAPTER 6 · PRACTICE

6.4 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

Do not divide only one term. In 3n+5=20, subtract 5 before dividing the whole side by 3.

GUIDED PRACTICE · FIRST STEPS

6.4.1 Solve n+7=15 and 4n=28. FLUENCY

6.4.2 Solve 3n+5=20 and check. FLUENCY

6.4.3 A border uses 4n+4 slabs. If 44 slabs are used, find the pond side n. APPLICATION

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

6.4.4 C=3+2d. Find d when C=19. APPLICATION

6.4.5 A student solves 2(x+3)=18 as x=6. Is this correct? Show a check. REASONING

6.4.6 Solve 5n−7=2n+11 and check. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 6 · EXPLORATION

The matchstick chain

Squares are joined in a line, sharing one side between neighbours. Count matches for 1, 2, 3, 4 and 5 squares.

> **INVESTIGATION**
> Criterion B. Build or draw stages, record a table, generalise and justify the rule.

SQUARES N	MATCHES	INCREASE
1	4	—
2		
3		
4		
5		

EXPLORATION · THE MATCHSTICK CHAIN

Criterion B. Build or draw stages, record a table, generalise and justify the rule.

B1 Complete the match counts for n=1 to 5. INVESTIGATION

B2 Write a term-to-term rule and a position-to-term rule. INVESTIGATION

B3 How many matches for 20 squares? Explain why counting one by one is inefficient. INVESTIGATION

B4 With 100 matches, what is the largest chain? Does 50 matches fit a complete chain? CHALLENGE

B5 Justify the rule from the structure, not only the table. INVESTIGATION

CHAPTER 6 · CHECKPOINT

Checkpoint and reflection

CHECKPOINT

Answer, then explain one result to a partner.

C1	Find the 10th term of 3n+1.	CHECK
C2	Evaluate 2a+3 for a=6.	CHECK
C3	Expand 4(x+2).	CHECK
C4	Solve 2n+3=13.	CHECK

Reflection

- Which representation made the idea clearest: a diagram, a table, symbols or words?
- Where did you check an answer instead of trusting it?
- Write one question you would ask a partner to test understanding.

CHAPTER 6 · APPLICATION

The courtyard pond

A square pond has side n metres. A one-metre-square slab border surrounds it. The outer square has side n+2.

REAL-LIFE APPLICATION

Criterion D. Simulated design. Slabs cost £3.50 each and delivery is £25 per order.

POND SIDE N (M)	1	2	3	4	5
Border slabs	8	12	16	20	24

APPLICATION · THE COURTYARD POND

Criterion D. Simulated design. Slabs cost £3.50 each and delivery is £25 per order.

D1 Find the border rule and the slabs for n=7 and n=12. APPLICATION

D2 A design uses 44 slabs. Find n and write the cost expression. APPLICATION

D3 With a £200 budget, find the largest whole-number pond side and check the next size. CHALLENGE

D4 Laying each slab takes 12 minutes, plus 2 hours preparation. Find the working time for the largest pond. APPLICATION

CHAPTER 6 · ASSESSMENT PRACTICE

The pattern explainer

MYP ASSESSMENT PRACTICE

Criteria A and C. Produce a short explanation with a diagram, a rule, a substitution and a check.

Explain the border rule 4n+4 to a learner who keeps counting the four corners twice.

ASSESSMENT PRACTICE · THE PATTERN EXPLAINER

Criteria A and C. Produce a short explanation with a diagram, a rule, a substitution and a check.

S1	Give two equivalent expressions for the border and explain one.	ASSESSMENT
S2	Use the rule for n=9 and check by drawing or area subtraction.	ASSESSMENT
S3	Write one misconception warning and a check question with answer.	ASSESSMENT

Success criteria

- Use correct mathematical language and notation.
- Show a method or representation that another student can follow.
- Check the result and explain why it is reasonable.

CHAPTER 6 · MIXED PRACTICE

Mixed practice 6

Connect ideas across the chapter. Give units and reasons where requested.

MIXED PRACTICE 6

Connect ideas. Give units and reasons where requested.

M1	Find the next two terms of 5, 9, 13, 17.	FLUENCY
M2	Evaluate $4n-3$ for $n=6$.	FLUENCY
M3	Simplify $5x+2-3x+7$.	FLUENCY
M4	Expand $2(3x-5)$.	FLUENCY
M5	Solve $5n=45$ and $n-8=12$.	FLUENCY
M6	Solve $3n+2=23$.	APPLICATION
M7	Explain why $2x+3$ cannot be simplified to $5x$.	REASONING
M8	A formula is $C=5+2n$. Find n when $C=31$.	APPLICATION
M9	Solve $4(n-2)=28$.	CHALLENGE
M10	Find the largest n with $3n+1 \leq 100$.	CHALLENGE

FINAL REFLECTION

Choose one item you solved confidently and one that needed a revision. Explain the difference in your method.

CHAPTER 6 · ANSWERS AND REASONING

Answers · Ready check and Checkpoint

READY CHECK

R1 **17.**
Multiply before adding.

R2 **14.**
Add 3.

R3 **28 and 7.**
Inverse operations.

R4 **5x.**
Add coefficients of like terms.

R5 **8.**
15−7.

CHECKPOINT

C1 **31.**
3×10+1.

C2 **15.**
12+3.

C3 **4x+8.**
Multiply both terms by 4.

C4 **n=5.**
Subtract 3, divide by 2.

CHAPTER 6 · ANSWERS AND REASONING

Answers · 6.1 and 6.2

PRACTICE 6.1 · PATTERNS

6.1.1 **16, 19; add 3.**
The difference is constant.

6.1.2 **3n + 1; 61.**
3×20 + 1.

6.1.3 **The first square needs 4; each extra square adds 3, giving $4+3(n-1)=3n+1$.**
Shared sides prevent counting four new matches for every square.

6.1.4 **4n + 4; 32 slabs.**
The outer square is $(n+2)^2$; subtract n^2.

6.1.5 **No. $3n+1=50$ gives $n=49/3$, not a whole number.**
The number of squares must be a positive integer.

6.1.6 **33 squares, using 100 matches exactly.**
$3n+1 \leq 100$; $n \leq 33$.

PRACTICE 6.2 · SUBSTITUTE

6.2.1 **17 and 35.**
Substitute, then multiply before adding.

6.2.2 **16.**
10+6.

6.2.3 **$x^2=4$; $-x^2=-4$.**
The first squares −2; the second applies the minus after squaring.

6.2.4 **3+2d; £19.**
3+2×8.

6.2.5 **$4n=12$, while $4+n=7$. Juxtaposition means multiplication.**
Substitution exposes the difference.

6.2.6 **$n=9$; $2\times9+7=25$.**
Subtract 7, then divide by 2.

CHAPTER 6 · ANSWERS AND REASONING

Answers · 6.3 and 6.4

PRACTICE 6.3 · SIMPLIFY AND EXPAND

6.3.1 **$6x-2$ and $5a+4$.**
Combine like terms only.

6.3.2 **$6x+12$ and $5y-10$.**
Multiply every inside term by the outside factor.

6.3.3 **$x+x=2x$. For $x=3$, this is 6, while $x^2=9$.**
Adding two copies differs from multiplying a number by itself.

6.3.4 **$2((2x+3)+4)=4x+14$.**
Perimeter is twice length plus width.

6.3.5 **No. Correct expansion is $3x+6$; at $x=1$, 9 is not equal to 5.**
The outside factor multiplies both terms.

6.3.6 **$6(2x+3)$.**
The HCF of 12 and 18 is 6.

PRACTICE 6.4 · SOLVE AND CHECK

6.4.1 **$n=8$ and $n=7$.**
Undo addition or multiplication.

6.4.2 **$n=5$; $3\times5+5=20$.**
Subtract 5, divide by 3, substitute.

6.4.3 **10 m.**
$4n+4=44$ gives $n=10$.

6.4.4 **$d=8$ km.**
$19-3=16$; $16\div2=8$.

6.4.5 **Yes. $x=6$ gives $2\times9=18$. An alternative method divides by 2 first: $x+3=9$, $x=6$.**
Both balanced methods agree.

6.4.6 **$n=6$; both sides equal 23.**
Subtract $2n$, add 7, divide by 3.

CHAPTER 6 · ANSWERS AND REASONING

Answers · Exploration

EXPLORATION · THE MATCHSTICK CHAIN

B1 **4, 7, 10, 13, 16.**
Each extra square adds 3 matches.

B2 **Start at 4 and add 3; matches = 3n+1.**
The first square has 4; each later square adds 3.

B3 **61. The rule gives the answer directly from the position.**
3×20+1.

B4 **33 squares uses 100 exactly. 50 does not fit: (50−1)/3 is not whole.**
Solve 3n+1 ≤ 100 and test divisibility by 3.

B5 **First square: 4 matches. Each additional square shares one side, so adds 3: 4+3(n−1)=3n+1.**
The shared-side argument applies to every stage.

CHAPTER 6 · ANSWERS AND REASONING

Answers · Application and Assessment

APPLICATION · THE COURTYARD POND

D1 **4n+4; 32 and 52 slabs.**
$(n+2)^2-n^2 = 4n+4$.

D2 **n=10 m. Cost = 3.5(4n+4)+25 = 14n+39 pounds.**
Solve 4n+4=44; multiply slabs by 3.50 and add delivery.

D3 **n=11 m costs £193; n=12 costs £207, so 11 is largest.**
$14n+39 \le 200$ gives $n \le 161/14 \approx 11.5$.

D4 **48 slabs take 9 h 36 min; plus 2 h gives 11 h 36 min.**
$4\times11+4=48$; $48\times12=576$ min=9 h 36 min.

ASSESSMENT PRACTICE · THE PATTERN EXPLAINER

S1 **$(n+2)^2-n^2$ and 4n+4. The first subtracts pond area from outer area; the second counts four sides plus four corners.**
Both count each border slab once.

S2 **40 slabs; $11^2-9^2 = 121-81 = 40$.**
A second representation checks the rule.

S3 **Sample: “Do not count corners twice: four sides of n slabs plus four corner slabs gives 4n+4.” Check n=3: 16 slabs.**
The warning names the error and the check verifies a case.

CHAPTER 6 · ANSWERS AND REASONING

Answers · Mixed practice

MIXED PRACTICE 6

M1 **21, 25.**
Add 4.

M2 **21.**
24−3.

M3 **2x+9.**
Combine like terms.

M4 **6x−10.**
Distribute the 2.

M5 **n=9 and n=20.**
Undo multiplication or subtraction.

M6 **n=7.**
Subtract 2, divide by 3.

M7 **2x and 3 are not like terms; at x=1 they give 5, while 5x gives 5, but at x=2 they give 7 and 10.**
Test a second value to expose the error.

M8 **13.**
(31−5)÷2.

M9 **n=9.**
Divide by 4, then add 2.

M10 **33.**
3n ≤ 99.